

Guide for Confidence and Predication Intervals from the Central Limit Theorem*

Eliza Oberstein

March 12, 2026

Contents

1	Central Limit Theorem	2
1.1	For Means	2
1.2	For Proportions	3
2	Confidence Intervals	4
2.1	Basic Recipe	4
2.2	Proportions	5
2.3	Means	5
2.3.1	One Mean	5
2.3.2	Two Means	5
2.4	Important things about Confidence Intervals	7
3	Confidence Intervals and Regression	8
3.1	CI for the slope	8
3.2	Confidence Intervals for Means	8
3.3	Confidence Intervals for Predictions	9

*This guide does not cover the whole course and may contain mistakes, if you see a mistake, please email me at eoerstei@ucsd.edu. Additionally, this will not be as detailed as your lecture notes but does aim to showcase a portion of the important material for the end of the course

1 Central Limit Theorem

1.1 For Means

The Central Limit Theorem says that when we are taking samples from a population with true mean μ and standard deviation σ . If we are taking a sample of $n \geq 30$ with random sampling and independence between responses. Independence can be assumed when we sample less than 10% of the total population.

When the previous assumptions are satisfied, we are able to approximate using a normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$

$$\boxed{\frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \sim_{\text{approx}} N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)}$$

The main thing we gain from this is that we can approximate a sample mean using the normal distribution, the results for the mean and standard deviation can be derived as follows

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} n\mu = \mu$$

Now, noting that we are assuming the X_i are independent we have

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2} n\sigma^2$$

Now we can take the square root to get

$$s.d.(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

1.2 For Proportions

A special application of the Central Limit Theorem is when we are working with proportions. In this case we are taking samples from a population and each response is either a yes (or 1) or a no (or 0)¹. In this case the setup will be a bit different.

We suppose that the true proportion in the population we are sampling from is p and we record $\hat{p} = \frac{\text{Number of Yes responses}}{\text{Total Responses}}$. Now, we again need to make sure we are sampling randomly and our samples are independent (which again can be assumed when $n \leq 10\%$ of the total population). And additionally, for the theorem to apply, we need $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

When the above assumptions are satisfied, we are able to use the approximation

$$\hat{p} \sim_{\text{approx}} N \left(p, \sqrt{\frac{p(1-p)}{n}} \right)$$

¹Note that this also applies to a binomial random variable when the same conditions are satisfied

2 Confidence Intervals

We now know that we have some approximate distribution for the sample means and proportions when the conditions for the CLT are met, and so we want to make some inferences about what the true population mean or proportion is. We can do this by the general principal of looking at how many standard deviations from the mean our sample would be. We know that because we have a normal or t distribution, we expect to be generally have observations within 3 standard deviations most of the time, but how can we be more precise.

Looking at the Central Limit Theorem, we note that we have multiple unknown parameters, the mean and the standard deviation, and so in order to construct a confidence interval we need to estimate the standard deviation in some way. We can do this by considering the standard deviation of the sample we are looking at, our approximation of the true sample statistic's standard deviation is called the standard error. When we have a mean, in order to account for this, the t distribution was introduced, and for proportions, we use the normal distribution.

2.1 Basic Recipe

In most situations we are able to follow the same basic steps and plug in the appropriate values.

Ingredients:

- Sample Statistic ($\bar{X}$ or $\hat{p}$)
- Standard Error ($s/\sqrt{n}$ or $\hat{p}(1 - \hat{p})\sqrt{n}$)
- Sample Size (n)
- Significance Level ($1-\alpha$)
- Critical Value (t^* or z^*)

Steps:

1. Calculate Sample Statistic
2. Use Significance Level to calculate Critical Value as follows

$\mathbb{P}(|Z| > z^*) = \alpha$ where $Z \sim N(0, 1)$ when using proportions, or

$\mathbb{P}(|T| > t^*) = \alpha$ where $T \sim t_{n-1}$ when using means²

3. Construct Our Margin of Error:

$$M.O.E. = \text{Critical Value} \times \text{Standard Error}$$

4. Construct the interval:

$$(\text{Sample Statistic} - M.O.E, \text{Sample Statistic} + M.O.E)$$

²Note that we can use $\mathbb{P}(N > z^*) = \frac{\alpha}{2}$ or $\mathbb{P}(T > t^*) = \frac{\alpha}{2}$ by symmetry.

2.2 Proportions

For proportions we follow the basic recipe as given using the normal distribution, this gives us

$$\left(\hat{p} \pm z^* \frac{\hat{p}(1 - \hat{p})}{\sqrt{n}} \right)$$

2.3 Means

2.3.1 One Mean

In this case our recipe becomes

Ingredients:

- Sample Statistic: $\bar{X}$
- Standard Error: s
- Sample Size: n
- Significance Level: $1 - \alpha$
- Critical Value: t^*

Steps:

1. Calculate Sample Statistic $\bar{X}$
2. Use Significance Level to calculate Critical Value as follows

$$\mathbb{P}(|T| > t^*) = \alpha \text{ where } T \sim t_{n-1}$$

3. Construct Our Margin of Error:

$$M.O.E. = t^* \times \frac{s}{\sqrt{n}}$$

4. Construct the interval:

$$(\bar{X} - M.O.E., \bar{X} + M.O.E.) = \left(\bar{X} \pm t^* \frac{s}{\sqrt{n}} \right)$$

2.3.2 Two Means

When we have two means we generally are making confidence intervals about their difference.

Paired Data: In this case we have two observations for each sample one belonging to each data set (ie math and verbal SAT scores for each student as you've seen in Labs) and we are able to find the sample standard deviation of the difference, in this case we use the exact same formula with

Test Statistic	$\frac{\bar{X}_d - \mu_d}{\frac{s_d}{\sqrt{n}}}$
Standard Error	$\frac{s_d}{\sqrt{n}}$
Degrees of freedom	$n - 1$

Which gives us a confidence interval of

$$\bar{X}_d \pm t^* \frac{s_d}{\sqrt{n}}$$

Independent Data: In this case we are considering the difference of two sample means $\bar{X}_1$ and $\bar{X}_2$ coming from two samples of size n_1 and n_2 with sample standard deviations s_1 and s_2 . And so when the conditions for the Central Limit Theorem are met we get that each is approximately normal and so if the sample means are independent, we know that the variance of the difference is the sum of the variances and the mean of the differences is the difference of the means.

$$\bar{X}_1 \sim_{\text{approx}} N(\mu_1, \frac{\sigma_1^2}{n_1})$$

$$\bar{X}_2 \sim_{\text{approx}} N(\mu_2, \frac{\sigma_2^2}{n_2})$$

$$\bar{X}_1 - \bar{X}_2 \sim_{\text{approx}} N(\mu_1 - \mu_2, \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}})$$

Now, we can approximate the standard deviation using the sample standard deviations and move to the t -distribution as before, and in this case the degrees of freedom is one less than the smaller of the two sample sizes. And so we find our critical value, depending on our significance level $1 - \alpha$ by finding t^* so that

$$\mathbb{P}(|T| > t^*) = 2\mathbb{P}(T > t^*) = \alpha \text{ where } T \sim t_{\min\{n_1, n_2\} - 1}$$

And now we can put our confidence interval together using the recipe to get

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

2.4 Important things about Confidence Intervals

There are a few sentences you may be asked about confidence intervals. There are two main things you should remember

Interpretation: The interpretation of a confidence interval is always going to be "We are (Confidence Level)% confident that the true population parameter is contained in our confidence interval". The only value that we can say we are confident is in our interval is the true population proportion, never any particular value.

If we made a 95% confidence interval for the proportion of people who were scared of roller coasters and found it to be $(0.35, 0.39)$, the interpretation would be "We are 95% confident that the true population proportion of people who are scared of roller coasters is between 35 and 39 percent"

If we made a 99% confidence interval for the mean number of minutes spent studying for the Math 11 final at UCSD and found it to be $(97, 118)$ we would say "We are 99% confident that the true population mean number of minutes spent studying for the Math 11 final at UCSD is between 97 and 118 minutes"

Confidence Level: What does 90% confident mean? It always means that "90% of intervals constructed in the same manor as ours will contain the true population proportion". This works with other percentages as well.

Width: You can see that the width of a confidence interval is the same as two times the margin of error, and so there are generally three factors that influence it, I will work with the MOE for means for the purposes of this, but it is similar for proportions: $MOE = t^* \frac{s}{\sqrt{n}}$. So there are three factors.

First is the standard deviation of the sample, we can see that as this gets bigger the interval gets wider and as it gets smaller the interval gets narrower.

Next we consider the confidence level, we remember that t^* is chosen so that $\mathbb{P}(T > t^*) = \frac{\alpha}{2}$ and our confidence level is $1 - \alpha$. An example of this is when we want a 90% confidence interval we choose t^* so that $\mathbb{P}(T > t^*) = 0.05$, ie only 5% of the data from the t_{n-1} distribution is greater than t^* . We can see that as we get more confident, we need to make sure t^* is bigger to cover more of the distribution, and so as we get more confident, the interval gets wider, and as we get less confident the interval gets narrower.

Finally, we see that the sample size impacts the width of the interval in two ways, first as we n increases, we get a $\sqrt{n}$ in the denominator, and so when we increase the sample size we will narrow the interval. Particularly, if we want to shrink the width in half, we have to quadruple the sample size as $\frac{1}{\sqrt{4n}} = \frac{1}{2} \frac{1}{\sqrt{n}}$ and to shrink it by a factor of 3 we need a sample 9 times as large. Additionally, as n increases, the t_n distribution gets squished closer in and t^* will decrease as well, which further narrow the interval. This does not exist when we are making intervals for proportions as the normal distribution does not depend on n .

3 Confidence Intervals and Regression

In the following section we will assume that two variables x and y which have the following true relationship

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Where β_0 is our intercept and β_1 is the slope of our regression line, and ϵ_i is an error. In practice we can make take for X and Y and we get a regression line $y = b_0 + b_1 x$, based on the actual data, and not the true relationship we have above. We can additionally estimate the residuals from the regression line $y_i - b_0 - b_1 x_i$.

Additionally, when we are doing confidence intervals for regression we lose an additional degree of freedom, ie we have $n - 2$ degrees of freedom.

In order to make confidence intervals, we need the following assumptions to be satisfied

- Linear Relationship between x and y : scatterplot appears linear
- Independence of ϵ_i : no patterns in residual plot
- Homoskedasticity of ϵ_i : residual plot appears to be evenly spread, no cone shapes
- Normality of ϵ_i : residual histogram appears normal

3.1 CI for the slope

When the above conditions are satisfied, we will often be given a slope coming from its sample and a corresponding standard error and so we can use the t_{n-2} distribution to get our critical value as before and we get a confidence interval of:

$$(b_0 \pm t^* S.E.(b_0))$$

3.2 Confidence Intervals for Means

Suppose we want to find a confidence interval for the mean of y when x is fixed to be x^* . In this case again we will be given a regression line and to get the center of our interval we will plug x^* into the regression line to get $\hat{\mu}_{y|x^*} = b_0 + b_1 x^*$, and we will likely be given a standard error and so we can make the confidence interval:

$$(\hat{\mu}_{y|x^*} \pm t^* S.E.(\hat{\mu}_{y|x^*}))$$

3.3 Confidence Intervals for Predictions

Sometimes, instead of making a confidence interval for the mean response for a fixed x value, we will have an individual and want to make a confidence interval for their prediction. The prediction is for $\hat{y}$ given some x^* and so we get the center of the interval to be $\hat{y} = b_0 + b_1x$ as before.

We know that there will be additional variance when we are considering just one person as opposed to a mean, and so our formula for the standard error will be different. But again, we are often given it. Now, we make the confidence interval as

$$\boxed{(\hat{y} \pm t^* S.E.(\hat{y}))}$$